Combined and Composite Games in Evolutionary Game Theory*

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Building upon previous work in analyzing finite population size effects on evolutionary games, defining new game as a compositon of well known 2 and 3 strategy games and derive replicator dynamics for different microscopic interaction processes. Computing drift and drift reversal (loss of stability below a critical population size) both through extensive simulations and analytical derivations for each of the agent interaction processes. Developed visualization tools.

I. INTRODUCTION: RPS DYNAMICS

A. Evolutionary game theory

Introduction to evolutionary game theory [1, 2]

To understand the combined game, first must look at the single cases of rock paper scissors and Snowdrift.

Report focusses mostly on math, derivations and results, for more information about the simulation code and visualisation tools, see project readme, and readme in /source/simulation for an example of how to use the library to run custom game simulations. Also how to use the gui to interact and run some animations and plots.

B. Rock-Paper-Scissors Model

Examples cyclic RPS dynamics appear in biology most notably, cyclic dominance in lizards [3], strains of e-coli [4]. RPS has been analyzed in the context of evolutionary game theory extensively [2]. Previous work focusses on the rock-paper-scissors game in finite populations and has shown that a drift reversal due to finite populations can arise under certain parameter conditions and microscopic interaction processes [5].

The standard rock paper scissors matrix is defined as: The combined game analysed mainly as part of this report is a composition of the rock-paper-scissors (RPS) game and the snowdrift (SD) game.

The general rock paper scissors payoff matrix is defined as:

$$\begin{bmatrix} a & c & b \\ b & a & c \\ c & b & a \end{bmatrix} \tag{1}$$

For the standard symmetric case of the RPS game, $a = 0, c = -1, b = 1$ where rock crushes scissors, paper covers rock, and scissors cuts paper. Ties are awarded the payoff of 0 resulting in cyclical dynamics about the internal fixed point of $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$.

1. Replicator dynamics

To analyze the time evolution of a population under the RPS game, the standard imitation based replicator equations can be derived for the system. Let x, y and z represent the fraction of players using rock, paper, and scissors respectively. The fraction of scissors players z can be defined as $z = 1 - x - y$, reducing the dimension of the replicator equations. Expected payoffs for each strategy can then be defined as follows:

$$\pi_R = ax + b\,(-x - y + 1) + cy \tag{2}$$
$$\pi_P = ay + bx + c\,(-x - y + 1) \tag{3}$$
$$\pi_S = a\,(1 - x - y) + by + cx \tag{4}$$

Using the standard replicator model for the evolution of population fractions over time, where the payoff of an individual is compared to the average payoff in the population, the following well known equation can be used to define a series of differential equations for each of the population fractions:

$$\dot{x}_i = x_i(\pi_i(x) - \langle \pi(x) \rangle) \tag{5}$$

where the average payoff (or mean fitness) in the population is:

$$\langle \pi(x) \rangle = x\pi_R + y\pi_P + z\pi_S \tag{6}$$

This leads to the two differential equations for the evolution of the population fractions x and y over time:

$$\begin{aligned} \dot{x} = {} & xy\,(ax - ay - bx + b\,(-x - y + 1) + cy - c\,(-x - y + 1)) \\ & + x\,(-x - y + 1)\,(ax - a\,(-x - y + 1) - by + b\,(-x - y + 1) - cx + cy) \end{aligned} \tag{7}$$

* Submitted for MSci final year project report, computer science

$$\begin{aligned}\dot{y} &= xy\left(-ax+ay+bx-b\left(-x-y+1\right)-cy+c\left(-x-y+1\right)\right)\\ &+y\left(-x-y+1\right)\left(ay-a\left(-x-y+1\right)+bx-by-cx+c\left(-x-y+1\right)\right)\end{aligned} \tag{8}$$

These are the determinisic payoff difference replicator equations from evolutionary game theory which define populations x, y over time. Later I will derive these for specific microscopic interaction processes including the noise terms that appear due to the finite population size effects.

The stability of this system of replicator equations can be analyzed by computing the eigenvalues of Jacobian of the replicator system, following standard methods in evolutionary game theory [2]:

$$J = \begin{bmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \\ \frac{\partial G}{\partial x} & \frac{\partial G}{\partial y} \end{bmatrix} \tag{9}$$

where $F = \dot{x}, G = \dot{y}$. To see full derivation of $\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial G}{\partial x}, \frac{\partial G}{\partial y}$ see appendix (standard RPS).

Fixed point stability can be determined by computing the eigenvalues of this matrix J at a specific point.

For the standard RPS game where $a = 0, b = 1, c = -1$:

$$\lambda_1 = -\frac{1}{2}\sqrt{16x^2+16xy-16x+16y^2-16y+4} \tag{10}$$

$$\lambda_2 = \frac{1}{2}\sqrt{16x^2+16xy-16x+16y^2-16y+4} \tag{11}$$

and and the internal fixed point of $\left(\frac{1}{3}, \frac{1}{3}\right)$ for RPS, this evaluates to

$$\lambda_1, \lambda_2 = \left\{-\frac{\sqrt{3}}{3}i, \frac{\sqrt{3}}{3}i\right\} \tag{12}$$

Since $\mathrm{Re}(\lambda_{1,2}) = 0$ and the eigenvalues are purely imaginary, the interior fixed point is a (nonlinear) center and the trajectories form closed periodic orbits at a fixed distance determined by the initial population fractions as seen in Fig. 1.

In this report I will focus on the case where $a = 0, b = 1$ and c is left as a free parameter which governs the dynamics of the game. In the standard case where $c = 1$, the system is neutrally stable and the trajectories oscillate at a fixed distance from the Nash equilibrium $(1/3, 1/3, 1/3)$. When $c < -1$ the system spirals outwards from the equilibrium and will be absorbed by the boundaries and finally when $c > -1$ the system will sprial inwards, where cooperation between the 3 strategies is preferred. It has previously been shown that a drift reversal effect occurs when $c > -1$ where a critical population size can reverse the drift from the internal stable point towards to boundaries [5].

With a population size of 500, the agent based simulation follows the deterministic trajectory rather well, with

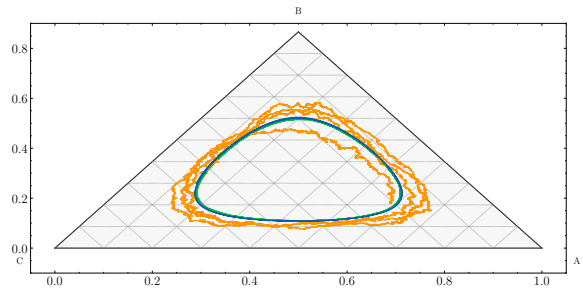

FIG. 1. Standard zero sum RPS game. $w = 0.3$, analytical trajectory (blue), large N (500000) in green and small N (2000) in orange for comparison. Large N closely follows the replicator dynamics as expected, small N is heavily influenced by noise and doesnt precisely follow dynamics. Agrees with stability analysis, periodic orbits at a fixed distance determined by the initial population.

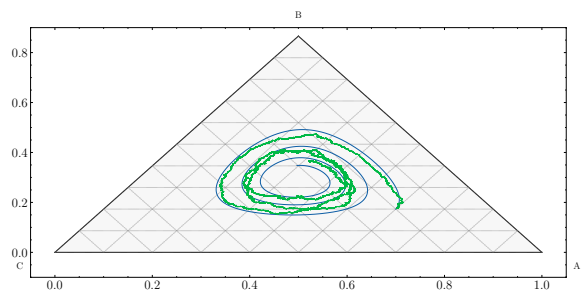

FIG. 2. Trajectory for the RPS game where $c = -0.8$. The blue line corresponds to a numerical solution of the adjusted replicator dynamics for the Moran process, and the green line is a Moran process agent based simulation with a population size of 500. Strength of selection $w = 0.3$

added noise due to the stochastic noise element arising from a finite population size (fokker planck equation).

The eigenvalues of the game in Fig. 2 are:

$$\lambda_1, \lambda_2 = \left\{-\frac{1}{30}+\frac{3\sqrt{3}}{10}i, -\frac{1}{30}-\frac{3\sqrt{3}}{10}i\right\} \tag{13}$$

Since the real part of the eigenvaleus are negative, $\mathrm{Re}(\lambda_{1,2}) = -\frac{1}{30} < 0$ the fixed point $\frac{1}{3}, \frac{1}{3}$ is linearly stable, so the population will converge towards the central fixed point over time. The imaginary component indicates the oscillatory dynamics so the population will spiral inwards towards the center (spiral sink). When $c = -0.8$, the populations converges towards a stable coexistence. For this particular case of RPS a drift reversal can occur in finite populations [5].

C. Snowdrift game

The snowdrift/hawk-dove game has also been intensely studied [1] and is an example of a anti-coordination game where unlike the famous prisoners dilemma, mutual cooperation is prefered over mutual defection. It describes situations where individuals compete over limited resources and the benefits and costs associated with them.

$$\begin{bmatrix} (\frac{b}{2}-\frac{c}{2},\frac{b}{2}-\frac{c}{2}) & (\frac{b}{2}-c,\frac{b}{2}) \\ (\frac{b}{2},\frac{b}{2}-c) & (0,0) \end{bmatrix} \tag{14}$$

The snowdrift game describes a scenario of two drivers trapped in a snowdrift. They have the options to shovel the snow an clear a path (cooperate) or stay in their car (defect). If both cooperate then they pay a shared cost $\frac{c}{2}$ but share the full benefit and can both drive. However the best outcome for one player is to remain in their car and let the other do all the work in turn leaving the full cost c to the other player but if both players remain in their car then both remain stranded and receive a payoff of 0. This scenario is often arises in biology and social/economic settings where mutual cooperation would be beneficial and both parties refusing to yield results in the worst outcome. For example in negotiation scenarios/research investing for competing companies/animals and microbes sharing resources in yeast cells [6] where some cooperating cells produce a necessary enzyme meanwhile "cheater" cells do not yet still receive the benefit they provide, other examples of cheating microbes [7].

Both the snowdrift and the hawk-dove game model a scenario where two players are competing over some resource but in a social/economic scenario and a biological competition however in both the cost of cooperation is high, and the cost of mutual defection is devastating. The resulting Nash equilibrium are two pure strategy equilibrium where players select opposite strategies (one gets the whole reward and no cost, whilst the other takes a smaller reward but incurs a heavy cost), and the mixed strategy $(\frac{1}{2},\frac{1}{2})$ between cooperate and defect, where players randomize their selection as defecting all the time results in a payoff of 0.

The mixed strategy case is typically stable [show plot and calculate eigenvalues for a typical example of the snowdrift game.]

II. COMBINED GAME DEFINITION

This work focusses on the composite game with the idea of combining snowdrift (SD) and rock-paper-scissors (RPS) into a single four strategy payoff matrix. Average payoffs, replicator equations and fixed points can subsequently be derived and the deterministic dynamics can be studied. The simulation tool can do all of this, plot trajectories both in 2d and and in the 3d pyramid simplex. These dynamics have been considered before [2], however not in extensive detail nor in finite populations. In this report I show that there is a drift reversal and that this occurs at different population sizes, and can be tuned to provide different distinct cases, altering the dynamics from the infinite N case depending on the relative critical population sizes.

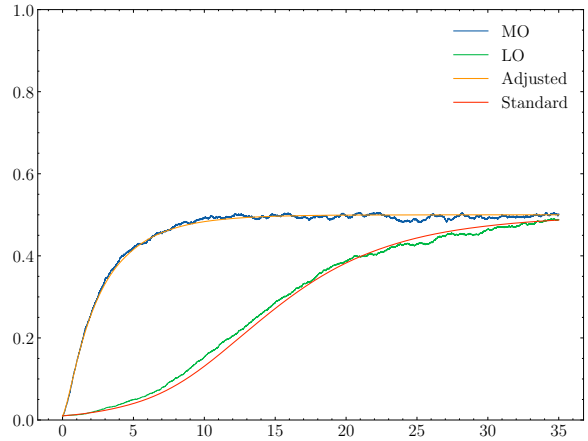

FIG. 3. Snowdrift game population time series for Local update (LU) and Moran process (MO), alongide solution to determinisic replicator equation for each process from the adjusted dynamics (ref fokker planck equation). $N = 10000$, large population size trajectories closely follow deterministic path. $w = 0.9$, initial population 99% cooperators. The game fixes at the mixed strategy Nash equilibrium $(\frac{1}{2},\frac{1}{2})$

The combined game is defined as:

$$\begin{bmatrix} a & c & b & \gamma \\ b & a & c & \gamma \\ c & b & a & \gamma \\ a+\beta & a+\beta & a+\beta & 0 \end{bmatrix} \tag{15}$$

where the values a, b, c represent the RPS cyclic game.

When $a = 0$, standard RPS ties, this can be seen in the 2x2 snowdrift context as the following payoff matrix:

$$\begin{bmatrix} \langle RPS \rangle & \gamma \\ \beta & 0 \end{bmatrix}$$

Where $\langle RPS \rangle$ is the expected value of the RPS game. The observation function H_{SD} equation 41 will measure the distance from the fixed point of this game purely in the SD context (the vertical column in the 4-simplex).

The population can be visualized as a 3d project of a 4-simplex with RPS at the base and the fourth (SD) strategy at the top. Cooperation in this game essentially collapses into playing RPS, whilst defection/cooperation pairs lead to typical snowdrift dynamics.

A. Replicator dynamics

The payoffs for this game are:

$$\begin{aligned}
\pi_R &= ax + bz + cy + \gamma(1 - x - y - z) \\
\pi_P &= ay + bx + cz + \gamma(1 - x - y - z) \\
\pi_S &= az + by + cx + \gamma(1 - x - y - z) \\
\pi_+ &= x(a + \beta) + y(a + \beta) + z(a + \beta)
\end{aligned}$$

Using + to denote the 4th strategy.

The standard replicator dynamics [2] Equation 5 considering the case where N is infinitely large:

$$\begin{aligned}
\dot{x} = {} & x(ax + bz + cy + \gamma(1 - x - y - z) - x(ax + bz + cy + \gamma(1 - x - y - z)) - y(ay + bx + cz + \gamma(1 - x - y - z)) \\
& - z(az + by + cx + \gamma(1 - x - y - z)) - (x(a + \beta) + y(a + \beta) + z(a + \beta))(1 - x - y - z))
\end{aligned}$$

$$\begin{aligned}
\dot{y} = {} & y\big(ay + bx + cz + \gamma(1 - x - y - z) - x(ax + bz + cy + \gamma(1 - x - y - z)) - y(ay + bx + cz + \gamma(1 - x - y - z)) \\
& - z(az + by + cx + \gamma(1 - x - y - z)) - (x(a + \beta) + y(a + \beta) + z(a + \beta))(1 - x - y - z)\big)
\end{aligned}$$

$$\begin{aligned}
\dot{z} = {} & z\big(az + by + cx + \gamma(1 - x - y - z) - x(ax + bz + cy + \gamma(1 - x - y - z)) - y(ay + bx + cz + \gamma(1 - x - y - z)) \\
& - z(az + by + cx + \gamma(1 - x - y - z)) - (x(a + \beta) + y(a + \beta) + z(a + \beta))(1 - x - y - z)\big)
\end{aligned}$$

B. Stability analysis

Defining $F = \dot{x}, G = \dot{y}, P = \dot{z}$ the Jacobian for this system is:

$$J = \begin{bmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z} \\ \frac{\partial G}{\partial x} & \frac{\partial G}{\partial y} & \frac{\partial G}{\partial z} \\ \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} & \frac{\partial P}{\partial z} \end{bmatrix} \tag{16}$$

Considering one particular instance of this game where $a = 0, b = 1, c = -1$ (standard rock paper scissors) and $\gamma = 0.2, \beta = 0.1$, there is an internal fixed point at $\left(\frac{2}{9}, \frac{2}{9}, \frac{2}{9}, \frac{1}{3}\right)$. The eigenvalues of this system are:

$$\lambda_1, \lambda_2, \lambda_3 = \left\{ -\frac{1}{15}, \frac{2\sqrt{3}}{9}i, -\frac{2\sqrt{3}}{9}i \right\} \tag{17}$$

The negative real eigenvalue here $-\frac{1}{15}$ indicates the stable plane where the trajectories will converge to (the snowdrift direction), and the pair of purely imaginary eigenvalues indicate the nonlinear center as before for stanrd rock paper scissors seen in Fig. 1. This means the system will converge to the fixed points RPS plane, and then oscillate at a fixed distance from the center point $\left(\frac{2}{9}, \frac{2}{9}, \frac{2}{9}, \frac{1}{3}\right)$. See the solved trajectory against simulated fininte popualtion trajectory in Fig. 4 and Fig. 5 for the normalized time series. Normalization is done for simulated results where simulated iterations are divided by the population size to match the time steps for the solved differential equations in this case 150 time steps which was sufficient to show the fixation at the stable orbit in the plane of the fixed point.

As for RPS, setting parameter $c = -0.8$ or any $c > -1$, the RPS system becomes attracting and the system would converge to the fixed points RPS plane, whilst also being attracted inwards to the fixed point indicating a stable coexistence of all four strategies. The eigenvalues in this case ($c = -0.8$) are:

[Add diagram and eigenvalues here for C = -0.8]

III. MODEL AND STOCHASTIC DYNAMICS

We consider a finite population of size N evolving under stochastic evolutionary dynamics with S strategies. The population state is described by the vector $\mathbf{x} = (x_1, \ldots, x_{S-1})$, where $x_i = n_i/N$ and the final strategy is fixed by normalization. At each update step, individuals interact according to a payoff matrix $\mathbf{A}$ and reproduce according to a specified microscopic update rule. In the large–N limit, the stochastic dynamics can be approximated by a Fokker–Planck equation governing the probability density $P(\mathbf{x}, t)$.

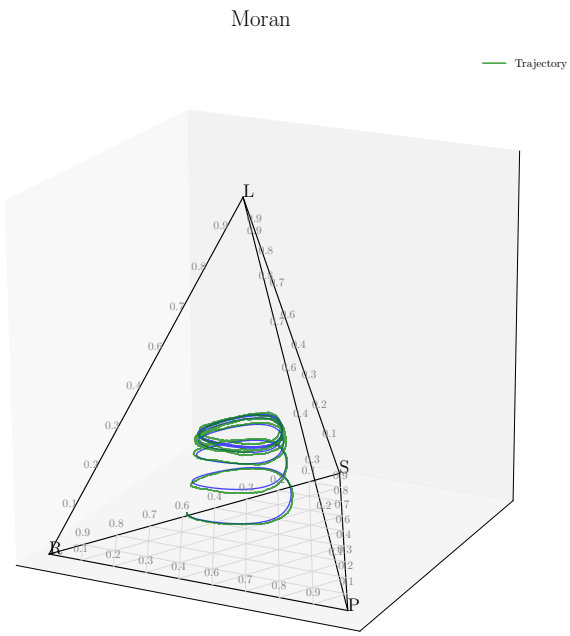

FIG. 4. Simplex trajectory, numerical / sim, w = 0.45, N = 50000, 150 time steps - normalized iterations N * 150 for simulation initial population [0.5,0.2,0.2, 0.1], RPS+, Standard RPS case corresponding to eigenvalues shown before $\gamma = 0.2, \beta = 0.1$, shows convergence to plane then periodic orbits at fixed distance from center (2/9, 2/9,2/9,1/3)

IV. INTERACTION PROCESSES

To study the dynamics in finite populations, an interaction process is required for comparison between selected agents at each time step.

A generalized form of the Moran process in terms of transition probabilites between states in the game simplex has been discussed in Ref.[8] and I have implemented in the simulation code for this project for any number of strategies (in this work focussing on 4 strategy model). Later these transition probabilities can be used to derive a master equation [9] which can be expanded in a Taylor series to derive an equation in the form of a Fokker-Planck equation describing the time evolution of the probability densities of the population states.

The Moran process models evolution as a birth death process where at each update step one individual is chosen proportional to their fitness in the population for reproduction, and one individual is chosen to be killed and replaced by the strategy of the chosen individual. For the selection process individuals need to have access to a global average population fitness as individuals are chosen relative to their fitness in the whole population.

In the context of the combined game the probability of switching from one strategy to another in the Moran process (specifically $R \to +$) is defined as:

$$T^{R\to+} = \frac{1}{2}\frac{1-w+w\pi_+}{1-w+w\langle\pi\rangle}\frac{i}{N}\frac{N-i-j-k}{N} \tag{18}$$

where $R, +$ denote rock and the 4th strategy and there is i rock players, and $N-i-j-k$ $(+)$ players.

And transitions within the RPS game will are of the form:

$$T^{R\to S} = \frac{1}{2}\frac{1-w+w\pi_S}{1-w+w\langle\pi\rangle}\frac{i}{N}\frac{k}{N} \tag{19}$$

where w is the strength of selection, how much the fitness of an individual plays a role in reproduction vs random selection. In weak selection limit where $w \to 0$, this transition probability just becomes $\frac{1}{2}\frac{N_i N_j}{N^2}$, and selection is purely random.

The transitions between the other strategies are just permutations of the above with the corresponding payoffs and populations. $T^{R\to+}$ will be simply written as T^{R+}, and similarly for other transitions from here.

Another model is the Local update process which considers two randomly selected individuals a, b in the population, and individual b transitions to individual a with a probability proportional to their difference in payoff. Unlike the Moran process this does not require every individual to know the global average payoff as reproduction is decided purely based on local interactions.

Individual b switches to a's strategy with probability:

$$p = \frac{1}{2} + \frac{w}{2}\frac{\pi_a - \pi_b}{\Delta\pi_{max}} \tag{20}$$

The transition probability of switching strategy is then:

$$T^{R\to+} = \left(\frac{1}{2} + \frac{w}{2}\frac{\pi_+ - \pi_R}{\Delta\pi_{\max}}\right)\frac{i}{N}\frac{N-i-j-k}{N} \tag{21}$$

Where $\Delta\pi_{max}$ is the maximum difference in payoff for the payoff matrix. In the standard RPS case where $\gamma, \beta < 1$, this is two as the maximum difference is the difference between an RPS win and loss.

The final model considered (and available in simulation library) is a local fermi process, similar to the local update process where only payoffs between two individuals are compared.

Probability of individual b taking a's strategy is:

$$p = \frac{1}{1+e^{-w(\pi_a-\pi_b)}} \tag{22}$$

And

$$T^{R\to+} = \left(\frac{1}{1+e^{-w(\pi_a-\pi_b)}}\right)\frac{i}{N}\frac{N-i-j-k}{N} \tag{23}$$

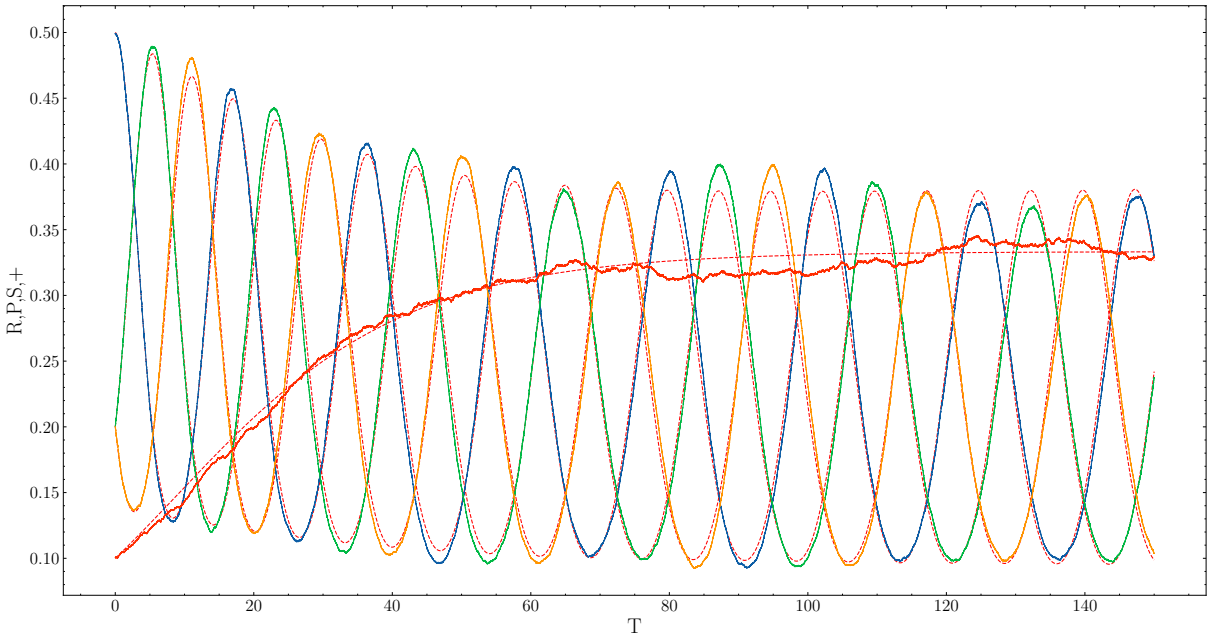

FIG. 5. Time series, for the same data plotted in Fig. 4, dashed line corresponds to the solved system of ODE's from the Moran adjusted dynamics, $a(x)$ from the Fokker-planck derivation.

A. General form

A general form for the transition probability between any two strategies can be given by:

$$\phi(b \to a)\frac{N_a N_b}{N^2} \tag{24}$$

Where a reproductive function ϕ is defined for each process. This is how it is implemented in augrps file, with functions to create a transition probs dict given a suitable reproduction function that meets the interface. (see appendix, and augrps for more detail in the symbolic implementation of this in python with sympy (reference sympy maybe?))

$$\phi_{MO}(\pi_a, \langle\pi\rangle, w) = \frac{1}{2}\frac{1 - w + w\pi_a}{1 - w + w\langle\pi\rangle} \tag{25}$$

$$\phi_{LU}(\pi_a, \pi_b, w) = \frac{1}{2} + \frac{w}{2}\frac{\pi_a - \pi_b}{\Delta\pi_{max}} \tag{26}$$

$$\phi_{FP}(\pi_a, \pi_b, w) = \frac{1}{1 + e^{-w(\pi_a - \pi_b)}} \tag{27}$$

V. FOKKER-PLANCK EQUATION

Master equation [9] can be derived and using the 2nd order taylor expansion of the transition probabilities for each strategy, a fokker-planck equation can be derived for each of the processes.

Used to describe particle motion (brownian motion) but can be applied to finite population with the defined hopping probabilities.

The master equation defines the time evolution of a probability density, and using the kramers-moyal expansion processes a FPE can be formed, a differential equation consisting of drift and noise, and in this case determinitic replicator dynamics, and stochastic noise from finite N.

The master equation is defined as:

$$P = \text{master eq here} \tag{28}$$

Expanding the probabilities and the transition probabilites to the second order taylor expansion, the master equation can be expanded

For full derivation of the FPE for this system, see appendix.

In the diffusion approximation, the master equation may be expanded to second order in $1/N$, yielding a Fokker-Planck equation of the form

$$\frac{d}{dt}P^\tau(x) = \frac{1}{N}\left[-\frac{d}{dx}(a(x)P^\tau(x)) + \frac{1}{2}\frac{d^2}{dx^2}\left(b^2(x)P^\tau(x)\right)\right] \tag{29}$$

Here $x \in (i, j, k)$ denotes the frequency of a given strategy in the RPS subsystem, and

$$a(i) = T^{PR} + T^{SR} + T^{+R} - T^{RP} - T^{RS} - T^{R+}$$

$$b(i) = \sqrt{\frac{1}{N}\left(T^{PR} + T^{SR} + T^{+R} + T^{RP} + T^{RS} + T^{R+}\right)}$$

for strategy i, rock and similarly $a(j), b(j)$ for paper and $a(k), b(k)$ for scissors can be defined in the same way with the pairs of transition probabilities into and out of the respective states and must be derived for each microscopic update process and set of payoff matrices. Full derivation of Fokker-Planck equation and expressions for $a(x), b(x)$ can be found in full report detailing 2nd order expansion of the master equation for the 4x4 game.

For each strategy, $a(x)$ corresponds to the deterministic replicator equation ($N \to \infty$), and $b(x)$ is the diffusion term corresponding to the stochastic drift. Code is capable of computing these values for each interaction process in order to plot the correct determinisic adjusted dynamics. For the local update process, this $a(x)$ is equal to the standard replicator payoff difference equation (with some scaling factor accounting for selection pressure w).

The standard replicator dynamics [2] correspond to the local update process, and matches the replicator equation derived from FP expansion, whereas the Moran process corresponds to the dynamics proposed by John Maynard Smith [1], where there is an additional normalization by the average payoff of the population. Maynard discussed whether the standard dynamics were too restrictive for cases where fitness is frequency dependent (in a finite population), and therefor proposed the adjusted dynamics where you can imagine an individual playing against the population as a whole via the average payoff term.

A langevin equation describing the trajectory of a single individual in the game, essentially a stochastic replicator equation follows from this:

$$\dot{x} = a(x) + b(x)\xi \tag{30}$$

When plotting a time series of a particular interaction process simulation, the determinisic dynamics (infinite N) comparison can be made by ignoring the noise term $b(x)$ as it contains a factor $1/N$, and solving numerically $a(x)$ which corresponds to the deterministic replicator dynamics for that interaction process.

A. Deterministic dynamics

Considering $a(x)$ for the different processes shows that for the Moran process, it corresponds to the adjusted replicator dynamics considered in Ref.[1] as shown in Ref.[10].

$$\dot{x}_{MO} = ... \tag{31}$$

The local update process corresponds to the standard replicator dynamics [2] used in section (reference rps introduction) to define the deterministic dynamics for RPS and SD but with an additional scaling by the constant selection pressure w which was not considerd in infinite N scenarios.

$$\dot{x}_{LU} = ... \tag{32}$$

$$\dot{x}_{FE} = ... \tag{33}$$

VI. TRAJECTORIES

This section will contain both numerical trajectories from the fokker p equation, and the simulated results from simulation engine. Also the link to all three processes and their deterministic limit in infinite N.

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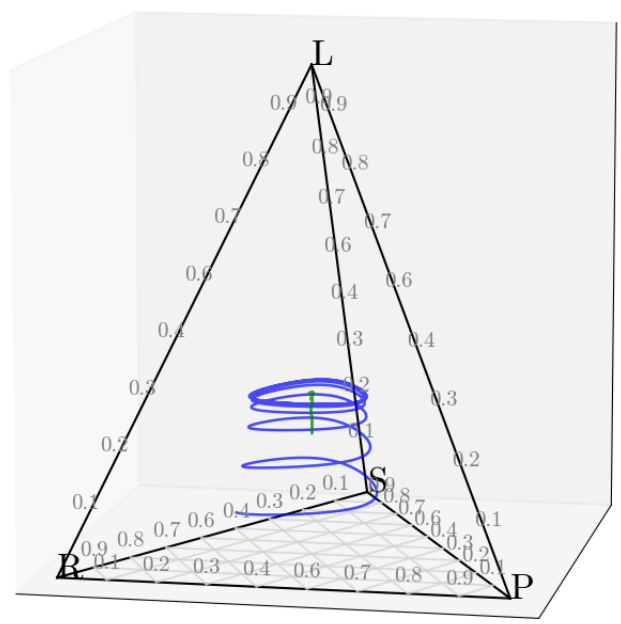

FIG. 6. Example of a trajectory with neutrally stable RPS component. The blue trajectcory corresponds to the numerically solved adjusted dynamics for the Moran process and the given payoffs. The interior green trajectory is an average trajectory for a population of 2000, averaged over 1000 independent simulation runs to show the general drift of the population in the SD component of the game. Note that drift is agreeing with replicator dynamics as $N_c < 2000$, this figure can probably be removed and replaced with one that shows drift reversal to the RPS

VII. DRIFT MEASURES

A measure of the drift from the internal fixed point of this system can be defined by an observation function H [5, 10, 11], and approximating the value of $\dot{H}$ with a discrete sum over the population. Previous work in Ref. [5, 10, 11] has examined this both the 2 and 3 strategy

case where only one observable H is used, however to analyze the combined game an observable will be defined for each of the sub games to properly describe the dynamics and drift. These will be H_{SD}, H_{RPS}, and H_4 as approximation of the distance from the SD equilibrium , the RPS equilibrium and then the overall central equilibrium as a whole respectively.

An approximation for $\dot{H}$, $\langle \Delta H \rangle$ can be computed directly from the agent based simulation, however for all the interaction processes the terms can be formulated as an integral and solved numerically on in some cases e.g. Local update, can be solved analytically. This is preferred over simulated approximate values as it is much simpler to compute particularly in the local update case where an exact result can be found for $\langle \Delta H \rangle$. [Put the graph here of critical N for simulated vs exact critical N.]

For each of the components making up the combined 4x4 game a suitable observation value H can be defined to measure the distance from the Nash equilibrium [5, 10]. The sign of $\dot{H}$ can therefore indicate whether the system is drifts towards or away from the interior equilibrium [10]. $\dot{H}$ is approximated in the simulated results by $\langle \Delta H \rangle$:

$$\langle \Delta H \rangle = \mathbb{E}[H(\mathbf{x}_{t+1}) - H(\mathbf{x}_t)]. \tag{34}$$

which is calculated for one single update step, and then averaged over a very large number of repeated simulations. Drift reversal is defined as the change in sign of this value under variation of parameters of the system most notably the population size N.

The sign of $\langle \Delta H \rangle$ determines whether stochastic drift favors interior coexistence or boundary attraction. $\langle \Delta H \rangle$ can be formulated analytically [5] and written as a sum over the simplex, in this case: $(1 \leq i, j \leq N - i, k \leq N - i - j)$.

$$\langle \Delta H \rangle = \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} (H_s - H_{s'})\, T^{s \to s'} \tag{35}$$

for all the states $s, s' \in \{R, P, S, L\}$ and transition probabilities $T^{s \to s'}$. Where transition probabilites $T^{s \to s'}$ must be defined for each of the microscopic interaction processes. In the combined SD, RPS game there are 3 observation values for drift, H_{SD} for effects within the snowdrift game, H_{RPS} for the RPS subgame and H_4 for the overall drift in the context of the internal fixed point of all 4 strategies in the payoff matrix.

After defining the summation in terms of discrete population counts for each of the strategies, it is possible to approximate these values in the continuous limit by replacing the counts with their corresponding fraction of the population and the term is converted to an integral over the simplex allowing for the use of numerical methods to compute these exact values of $\langle \Delta H \rangle$, and therefore compute numerically the critical population sizes N where the dynamics shift and drift reversal occurs (if it does occur). This allows for more efficient comparisons of the dynamics of a particular instance of the game parameters as it is much more efficient to compute this equation numerically than to simulate over a large number of realizations and average the discrete results.

A. Rock-Paper-Scissors Drift

For the three-strategy subsystem, the replicator equations have the constant of motion:

$$H_{\text{RPS}} = -xyz \tag{36}$$

$\langle \Delta H_{RPS} \rangle$ is approximated by a sum:

$$\begin{aligned}
\langle \Delta H_{RPS} \rangle = \frac{2}{N^6} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijk(T^{RP} + T^{RS} + T^{R+} \\
& + T^{PR} + T^{PS} + T^{P+} + T^{SR} + T^{SP} + T^{S+} \\
& + T^{+R} + T^{+P} + T^{+S}) - k(i-1)(j+1)T^{RP} \\
& - (i-1)j(k+1)T^{RS} - jk(i-1)T^{R+} \\
& - k(i+1)(j-1)T^{PR} - i(j-1)(k+1)T^{PS} \\
& - ik(j-1)T^{P+} - (i+1)j(k-1)T^{SR} \\
& - i(j+1)(k-1)T^{SP} - ij(k-1)T^{S+} \\
& - jk(i+1)T^{+R} - ik(j+1)T^{+P} \\
& - ij(k+1)T^{+S} \Big]
\end{aligned} \tag{37}$$

$$\begin{aligned}
\langle \Delta H_{\text{RPS}} \rangle = & \frac{2}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\
& \Big[z(y-x)(T^{RP} - T^{PR}) \\
& + y(z-x)(T^{RS} - T^{SR}) \\
& + x(y-z)(T^{SP} - T^{PS}) \\
& + xz(T^{P+} - T^{+P}) + yz(T^{R+} - T^{+R}) \\
& + xy(T^{S+} - T^{+S}) \Big] \\
& + \frac{2}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\
& \Big[z(T^{RP} + T^{PR}) + y(T^{RS} + T^{SR}) \\
& + x(T^{SP} + T^{PS}) \Big]
\end{aligned} \tag{38}$$

The RPS drift is therefore controlled by differences of transition probabilites between the RPS states as in [5] with additional terms in and out of the RPS game itself to the 4th strategy.

B. Snowdrift Drift

The 4th strategy occurs with frequency $q = 1-x-y-z$, we define the snowdrift coexistence measure

$$H_{SD} = -q(1-q) \tag{39}$$

The expected value of ΔH_{SD} over the entire simplex can be defined, similarly to the RPS case, and approximated by a sum as follows:

$$\begin{aligned}\langle \Delta H_{\mathrm{SD}} \rangle = \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& (N-i-j-k)(1-N+i+j+k)(T^{R+} + T^{P+} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) \\ & - (N-i-j-k+1)(-N+i+j+k)T^{R+} - (N-i-j-k+1)(-N+i+j+k)T^{P+} \\ & - (N-i-j-k-1)(2-N+i+j+k)T^{+R} - (N-i-j-k-1)(2-N+i+j+k)T^{+P} \\ & - (N-i-j-k-1)(2-N+i+j+k)T^{+S} \Big]\end{aligned} \tag{40}$$

$$\begin{aligned}\langle \Delta H_{\mathrm{SD}} \rangle = & \frac{12}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\ & \Big[q \big[(T^{R+} - T^{+R}) + (T^{P+} - T^{+P}) \\ & + (T^{S+} - T^{+S}) \big] \Big] \\ & + \frac{12}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\ & \Big[T^{+R} + T^{+P} + T^{+S} \Big]\end{aligned} \tag{41}$$

Since transitions purely within the RPS subgame do not change the value of H_{SD}, this exact term for depends only on transitions into and out of the fourth strategy (+).

Consider the case of $(H_{SD} - H'_{SD})T^{RP}$ in the summation being zero, since transitions between rock and paper do not change the frequency of the fourth strategy, and therefore do not change the value of $-q(1-q)$.

This results in $H_{SD} - H'_{SD} = H_{SD} - H_{SD} = 0$ for all transitions ($R \to P, P \to S, S \to R$, and their reverses) within the RPS subgame. For full derivation see the full report.

C. Four-Strategy Coexistence Drift

To capture full four-strategy coexistence, the constant of motion H_4 is defined

$$H_4 = -xyzq \tag{42}$$

which vanishes on all faces of the simplex and is maximized in the interior.

The expected value of ΔH_4 can be defined as the sum over all the states below

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N^7} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijk(N-i-j-k)(T^{RP} + T^{RS} + T^{R+} + T^{PR} + T^{PS} + T^{P+} \\ & + T^{SR} + T^{SP} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) - (i-1)(j+1)k(N-i-j-k)T^{RP} \\ & - (i-1)j(k+1)(N-i-j-k)T^{RS} - (i-1)jk(N-i-j-k+1)T^{R+} \\ & - (i+1)(j-1)k(N-i-j-k)T^{PR} - i(j-1)(k+1)(N-i-j-k)T^{PS} \\ & - i(j-1)k(N-i-j-k+1)T^{P+} - (i+1)j(k-1)(N-i-j-k)T^{SR} \\ & - i(j+1)(k-1)(N-i-j-k)T^{SP} - ij(k-1)(N-i-j-k+1)T^{S+} \\ & - (i+1)jk(N-1-i-j-k)T^{+R} - i(j+1)k(N-1-i-j-k)T^{+P} - ij(k+1)(N-1-i-j-k)T^{+S} \Big]\end{aligned} \tag{43}$$

In the continuous limit

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& zq(y-x)(T^{RP}-T^{PR}) + yq(z-x)(T^{RS}-T^{SR}) + xq(y-z)(T^{SP}-T^{PS}) \\ & + yz(q-x)(T^{R+}-T^{+R}) + xz(q-y)(T^{P+}-T^{+P}) + xy(q-z)(T^{S+}-T^{+S})\Big] \\ & + \frac{6}{N^2}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[zq(T^{RP}+T^{PR}) + yq(T^{RS}+T^{SR}) + xq(T^{SP}+T^{PS}) \\ & + yz(T^{R+}+T^{+R}) + xz(T^{P+}+T^{+P}) + xy(T^{S+}+T^{+S})\Big]\end{aligned} \tag{44}$$

This expression naturally decomposes into RPS–like drift terms, snowdrift–like drift terms, and diffusion corrections. The sign of $\langle \Delta H_4 \rangle$ determines whether stochastic dynamics favors four–strategy coexistence or boundary attraction.

D. Weak selection limit

Value of these integrals as $w \to 0$

E. Local update analytical solutions

For the Local update process, an exact solution for the values $\langle \Delta H_{SD} \rangle$ and $\langle \Delta H_{RPS} \rangle$ can be found.

Considering the payoff matrix:

$$\begin{bmatrix} 0 & c & 1 & \gamma \\ 1 & 0 & c & \gamma \\ c & 1 & 0 & \gamma \\ \beta & \beta & \beta & 0 \end{bmatrix} \tag{45}$$

In this case the values of a, b have been set to 0 and 1. The value of c can be altered to adjust the dynamics of the RPS component of the game, if $c > -1$ then the game will spiral inwards towards a central fixed point and if $c > -1$ the RPS trajectories will sprial outwards as shown in the introductory section. The values of β and γ determine the dynamics of the snowdrift component of the game with the main cases being fixation at either the top (pure 4th strategy), base (pure RPS) or a central equilibrium. This depends on the relative payoff difference between the SD and RPS game.

For the full derivation of these exact solutions, see appendix section (analyical solution for the local update process)

$$\langle \Delta H_{RPS} \rangle_{LU} = \frac{1}{120N^2} - \frac{w(2+\gamma-3\beta=2c)}{3360N\Delta\pi_{max}} \tag{46}$$

Setting this to 0, an exact solution for the local update critical population size is:

$$N'_{RPS} = \frac{28\Delta\pi_{max}}{w(2+\gamma-3\beta+2c)} \tag{47}$$

Similarly for the SD observable:

F. Comparison of drift measures

Drift reversal, below a critical population size, occurs at the point where the value of $\langle H \rangle$ changes sign. Depending on the initial parameters, this effect is observed in the RPS and SD subgames, and in some cases in both in the agent based simulations and also from the exact solutions for $\langle \Delta H \rangle$ for the different interaction processes and in the case of the local update process, exact expressions for the critical population size for a special case of the game are derived in section VII E.

In either cases this leads to a few notable cases resulting in very different dynamics in the multiple game system summarized by Fig. 7

Assuming a set of parameters where both a critical population size exists for both the SD and RPS part of the game where N'_{SD}, N'_{RPS}, N, are the critical sizes and the actual population size respectively there are the cases:

1. $N'_{SD} < N < N'_{RPS}$
2. $N'_{RPS} < N < N'_{SD}$
3. $N'_{RPS} \approx N \approx N'_{SD}$
4. $N < N'_{RPS}, N'_{SD}$

In case 1, population size is below the critical size for drift reversal in the RPS component, and therefore (for an attracting RPS game where $c > -1$) the trajectories would gradually move away from the internal fixed point in the RPS plane on average where in the large N case they would fixate there. Since N is larger than the critical size for the SD component there would be no reversal in the SD game and the trajectories would follow the drift expected from the replicator determinstic dynamics, e.g. the center of the vertical column in the 3d 4-simplex projection (for the setting where there is stable mixed strategy nash equilibrium in SD). This corresponds to the case in the bottom right of Fig. 7, where RPS trajectories fill out the plane and eventually in time are absorbed by the RPS boundaries.

In the case where $N'_{RPS} < N < N'_{SD}$ the point cloud of trajectories from randomly initialized simulations will be attracted to the equilibrium of the RPS game, but

will exhibit a drift reversal in the SD game, seen in the SD reversal case in Fig. 7.

Case 2 is the exact opposite of this where a reversal takes place in the SD component and not the RPS component resulting in the cloud of randomly initialised trajectories to fill out a vertical column in the 3d projection as they diverge from the expected central fixed point for the given payoff matrix (see inset). Here since $N > N'_{RPS}$ there is no reversal in the RPS component and since this RPS game is attracting and mutual coexistence is optimal, the trajectories remain in the center of the RPS plane meanwhile the SD components dynamics reverse.

[Table here of the exact values from the appendix.]

VIII. CONCLUSION

Discussion of the cases,

idea of a central blob like case where the Ns are all equal and a lower N than the large N case.

While $\langle \Delta H_{\mathrm{RPS}} \rangle$ and $\langle \Delta H_{\mathrm{SD}} \rangle$ describe drift within lower–dimensional subspaces, $\langle \Delta H_4 \rangle$ captures stability of full interior coexistence. Sign reversals of $\langle \Delta H_4 \rangle$ generally occur at larger population sizes, reflecting the additional geometric suppression of interior states in the four–strategy simplex.

Eigenvalues for each of the games are:

table here with 3 eigenvalues for each of the payoff matrices for the large diagram.

briefly discuss stability and reference the drift reversal in each case SD, RPS e.g attracting fixed point becomes repelling in RPS, but not SD

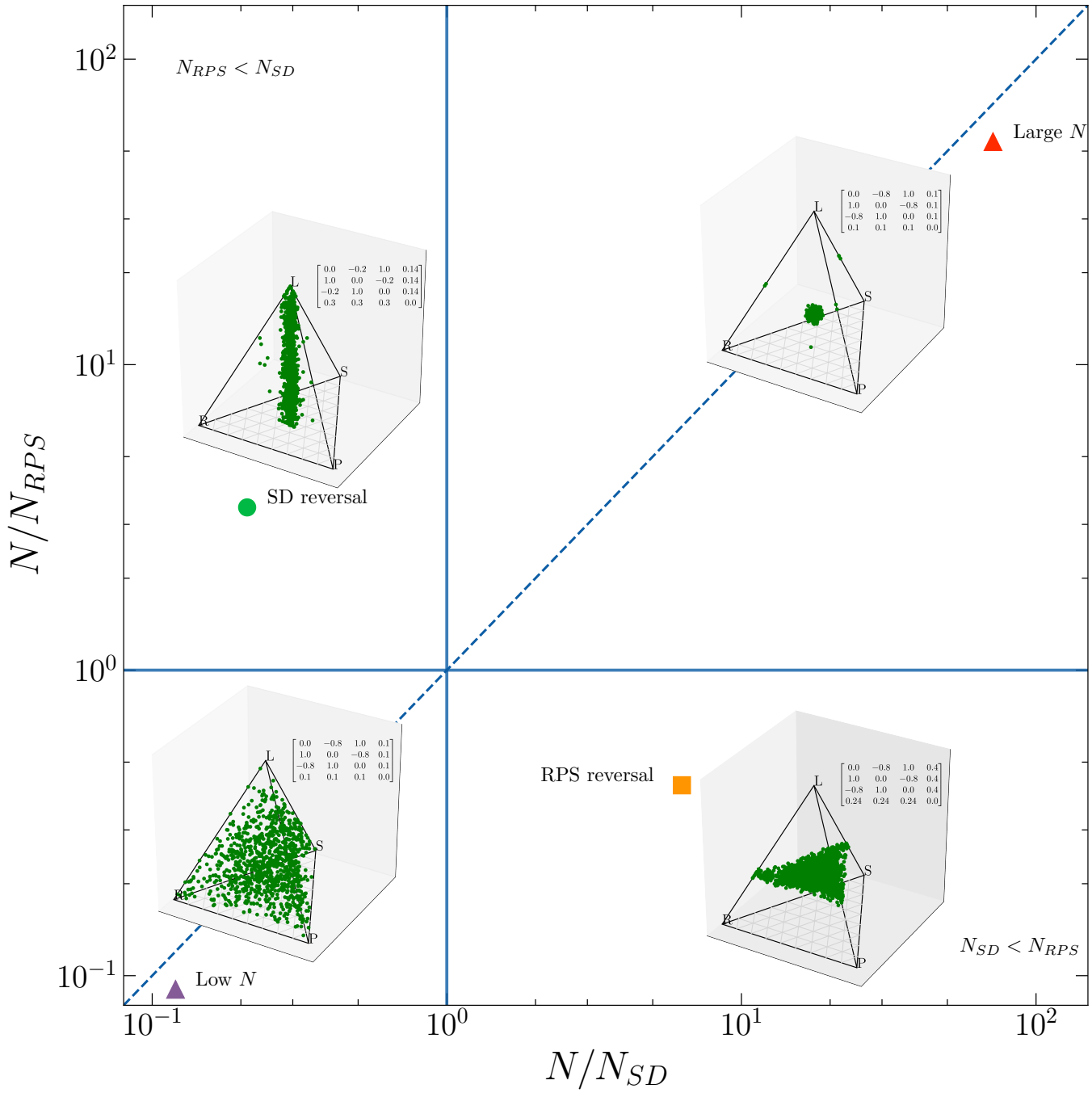

FIG. 7. Comparison of the main drift reversal cases, notably the RPS and not SD reversal (disk like shape), SD and not RPS reversal (column shape), and large N Nash equilibrium stable case where $N >> N_c$ and chaotic low N case (drift in all directions) where $N << N_{RPS}, N << N_{SD}$. Each point corresponds to an independent trajectory which started from an initial random point in the simplex.

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